Luxembourg: energy security, economic opportunity and partnership priorities

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AI-generated from the data and checked against it; paper baseline as of 12 October 2026.

Summary

- Luxembourg is most exposed in upstream nuclear, upstream oil and upstream batteries, where its Energy Security is higher than only 8%, 8% and 11% of the other economies scored. In upstream oil, no recorded reserves and a trade deficit pull it down.[1]
- Its strongest openings are in downstream geothermal, downstream green hydrogen and downstream batteries, where its Economic Opportunity is higher than 93%, 89% and 88% of the other economies scored; consumption pushes them up.[1]
- On the paper's weighting, the Strategic Index puts downstream solar 1st among Luxembourg's own sectors, almost tied with upstream nuclear. Data-based priority and policy activity are both high in upstream batteries, midstream nuclear and midstream batteries.[1]
- The Partnership Strength Index rates the United States highest for Luxembourg, as a reliable supplier. It is also Luxembourg's 8th trade partner; Germany, its 1st, ranks 3rd.[1,2]

1. Energy security risks and why

19 of Luxembourg's 30 placed sectors sit in the two low-energy-security corners, where its Energy Security is below the middle economy's. Energy Security measures how reliably a country can meet its needs in a sector: production, reserves, imports, the concentration of world supply and prices.[1]

Compared with other economies, its Energy Security is lowest in upstream nuclear, upstream oil, upstream batteries and downstream solar. In upstream nuclear its score is 0.16, higher than only 8% of the other 174 economies scored. The scores are 0.36 in upstream oil (higher than 8% of 96) and 0.17 in upstream batteries (higher than 11% of 76). With Economic Opportunity also below the middle, these upstream sectors sit in the Secure corner, which the paper's framework associates with securing supply through trusted partners. In downstream solar, its score of 0.42 is higher than 12% of the other 147, in the Upgrade corner of low Energy Security and high Economic Opportunity.[1]

Country Insights takes the high-priority sectors from the Upgrade corner in Strategic Index order: downstream solar, midstream solar and midstream nuclear. In downstream solar, Energy Security is low mainly because of its solar resource and energy access. Its score on solar resource potential is 0.15 against a median of 0.54. Consumption per person in the sector rose by 72.6% in the years to 2025, a measure on which it scores higher than only 10% of the other economies (higher means more secure).[1,3]

In upstream oil, the score is low mainly because of reserves and trade. No proved oil reserves are recorded, which the index counts as zero, and its trade deficit in these goods was 2.2% of GDP in 2023. For context, its largest suppliers in 2024 were Belgium (84.7% of imports) and Germany (9.9%).[2,4]

2. Economic opportunities and why

18 of the 30 placed sectors sit in the two high-opportunity corners. Economic Opportunity measures how well placed a country is to create value in a sector, from exports, related capabilities, investment, costs, demand and technology readiness.[1]

Its Economic Opportunity is highest in downstream geothermal, downstream green hydrogen, downstream batteries and upstream green hydrogen, where it scores higher than 93%, 89%, 88% and 84% of the other economies scored, respectively. In downstream geothermal and upstream green hydrogen, Energy Security is below the middle, so both sit in the Upgrade corner.[1]

Cost competitiveness pushes the score up in downstream geothermal, downstream batteries and upstream green hydrogen. On input costs for labour and capital, Luxembourg scores higher than 86%, 89% and 87% of the other economies there (higher means more opportunity). In downstream green hydrogen, consumption pushes it up: on expected growth in use per person it scores higher than 81% of the other economies. Part of the score in downstream geothermal rests on filled-in data: 46.7% of its weight.[1]

In midstream oil, its 5th highest sector on Economic Opportunity relative to other economies, trade pushes the score up. On export feasibility, which measures how close these goods are to what it already exports well, it scores higher than 76% of the other economies. For context, its largest markets in 2024 were Suriname (30.7%) and Guyana (24.7%).[1,2]

3. Sectoral priorities and the existing policy posture

The Strategic Index ranks a country's own sectors by opportunity, the shortfall in Energy Security, policy activity, stage and low-emission value. On the paper's weighting, it ranks downstream solar 1st (0.67), upstream nuclear 2nd (0.66), upstream batteries 3rd, upstream wind 4th and midstream batteries 5th among Luxembourg's own sectors. The scores of downstream solar and upstream nuclear are almost tied, 0.005 apart, as are those of upstream nuclear and upstream batteries (just under 0.010) and of upstream wind and midstream batteries (0.003).[1]

The lead of downstream solar rests on policy activity: its policy term adds 0.200, against 0.105 for upstream nuclear. Without that term, downstream solar falls in the middle third of its sectors, while upstream nuclear, upstream batteries and upstream wind are in the top third.[1]

In the allied network design, Luxembourg's pick is midstream batteries, almost tied with midstream solar (0.001 ahead). Higher-ranked upstream sectors are screened out because no resource base is recorded for them.[5]

Country Insights splits data-based priority (the Strategic Index without its policy term) and policy activity into thirds; both are high in upstream batteries, midstream nuclear and midstream batteries. No sector pairs high data-based priority with the least policy, or low priority with the most. Luxembourg carries the EU's Climate Action Tracker rating, Insufficient (November 2024), and has 130 International Energy Agency policy records, 61 of which name a technology. Country Insights also lists Global Trade Alert's industrial-policy measures recorded under Luxembourg, including EU-wide ones, as context; the policy term does not count them.[1,6,7]

4. Partnership priorities and why

The Partnership Strength Index rates partners for Luxembourg as reliable suppliers (Friendshore), as export markets (Partnership Opportunity) and, for middle- and lower-income partners, by Development Potential.

It rates the United States highest for Luxembourg, standing out most as a reliable supplier; its best five supplier sectors are upstream wind, downstream geothermal, upstream gas, upstream oil and downstream gas. It ranks Belgium 2nd, Germany 3rd, the Netherlands 4th and Italy 5th. Germany stands out as a reliable supplier, Belgium as an export market, the Netherlands as an export market and Italy as an export market.[1]

Each of these partners is a priority partner with strong ties today. Germany is Luxembourg's 1st trade partner on average across its energy-goods baskets, Belgium its 2nd, the Netherlands its 4th, Italy its 5th and the United States its 8th. Each shares membership of NATO with Luxembourg and sits in the same bloc in UN votes. By contrast, China is a strong tie today that the index ranks lower: its 9th trade partner, ranked 46th by Partnership Strength. In UN votes, the pair falls in opposite blocs.[1,2,8,9]

As a partner to the coalition, Luxembourg ranks 27th of 192 economies. In the allied network design, its derived shares include 0.1% of designed production in midstream batteries and 0.1% in midstream green hydrogen.[1,5]

5. Data notes and caveats

Paper baseline as of 12 October 2026; the Country Insights page holds the detail.

- Scores compare Luxembourg with other economies in the same sector; they are not measures of risk or profit in absolute terms.
- All index values are the paper's baseline as of 12 October 2026; Strategic Insights recomputes them under other weightings.
- Some figures behind these scores are filled in where the source has no value: with zero where a source lists every holder, otherwise with the world average.
- Strategic Index scores within 0.01 of each other are almost tied: small changes in the data or the weights can swap their order.
- Many allied picks are close calls: in the paper, 37% of them are within 0.01 of the runner-up.
- In the allied network design the European Union is one producer; Luxembourg's shares are derived from its weight within the EU.
- The Climate Action Tracker rates the EU as a whole; Luxembourg carries the EU's rating.
- Luxembourg's counts leave out the EU's own 438 International Energy Agency policy records.
- Trade figures count trade with other EU members.

Sources

1. Energy Security, Economic Opportunity, Strategic Index and Partnership Strength Index, paper baseline (Opportunity Security Indices pipeline); CC BY 4.0 (the paper's indices).
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3. Energy Institute, Statistical Review of World Energy 2026 (data to 2025); attribution; https://www.energyinst.org/statistical-review.
4. Growth Lab at Harvard University, The Atlas of Economic Complexity, HS92 6-digit country-product-year data (data to 2023); attribution; https://atlas.hks.harvard.edu/.
5. Allied network design (the paper's model design of coalition production shares).
6. Climate Action Tracker, country ratings (ratings updated 2022-2025); attribution; https://climateactiontracker.org/.
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8. Economic-security agreements: G7, NATO, Indo-Pacific Economic Framework, Minerals Security Partnership (and Forum), U.S. free-trade agreements (compiled by the authors from official membership lists).
9. Bailey, M., Strezhnev, A. and Voeten, E., United Nations General Assembly Voting Data, ideal points (Harvard Dataverse V39.0) (sessions to 2023); attribution; https://dataverse.harvard.edu/dataset.xhtml?persistentId=hdl:1902.1/12379.

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